

Week 3 BSM Chooser Option Validation

This report contains the quantitative analysis of the Rubinstein Chooser Option validation, incorporating analytical Greeks sensitivity, multi-scale Monte Carlo Path convergence checking, and choice time t convergence studies.

Core Model Equations

Split-Leg Model (Codebase Baseline):

$$V_{\text{split}} = C(S_0, K, T) + P(S_0, K e^{-r(T-t)}, T-t)$$

Rubinstein Exact Valuation (Standard Derivative Theory):

$$V_{\text{rubinstein}} = C(S_0, K, T) + e^{-q(T-t)} P(S_0, K e^{-(r-q)(T-t)}, t)$$

Table 2 replication setup

- Spot price: 156.7
- Strike price: 150.0
- Risk-free rate: 0.0015
- Dividend yield: 0.0233
- Volatility: 0.282
- Decision time t: 0.5 Years
- Maturity T: 1.0 Years
- Price drift used in the GBM path: $r - q = -0.0218$
- Stochastic seed: 17170

Base Valuation Breakdown

metric	value
spot_price	156.700000
strike_price	150
adjusted_strike_for_put_leg	149.887542
risk_free_rate	0.001500
dividend_yield	0.023300
volatility	0.282000
decision_time_years	0.500000
maturity_years	1
time_to_decision_years	0.500000
call_leg_value	18.688997
put_leg_value	9.713217

metric	value
chooser_value	28.402215
chooser_premium_vs_call	9.713217
chooser_premium_vs_put	18.688997
chooser_identity_gap	0

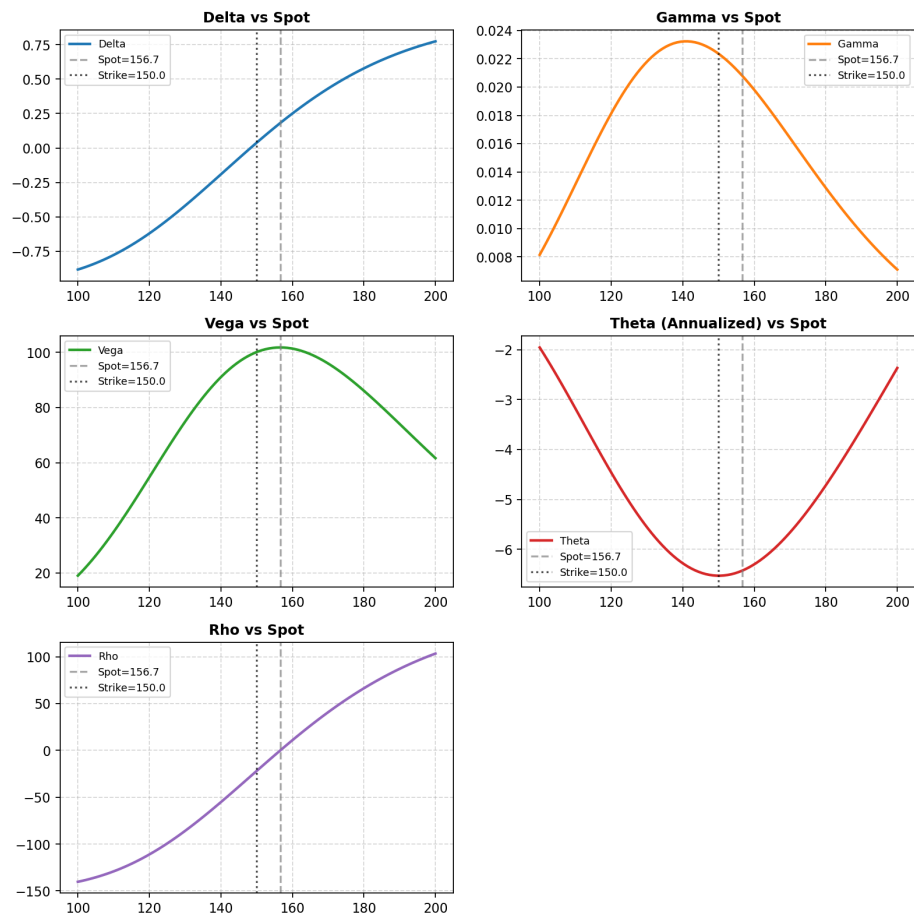
Part 1: Option Greeks & Sensitivity Analysis

Analytical calculation of the chooser option Greeks handles the decomposition of both the call and put legs, which is verified below against spot change curves. For the Rho metric, the derivative with respect to the risk-free rate includes the chain rule for the put strike parameter: $dK'/dr = -(T-t) K'$, leading to an exact analytical adjustment: $dP/dr_{total} = 2 * dP_{merton}/dr$.

Metric	Call Leg	Put Leg	Chooser Option
Delta (dy/dS)	0.573031	-0.389804	0.183227
Gamma (d2y/dS2)	0.008612	0.012175	0.020786
Vega (dy/dv)	59.631877	42.151674	101.783551
Theta (dy/dt)	-6.422553	0	-6.422764
Rho (dy/dr)	71.104965	-70.795501	0.309464

Greeks Sensitivity to Asset Price

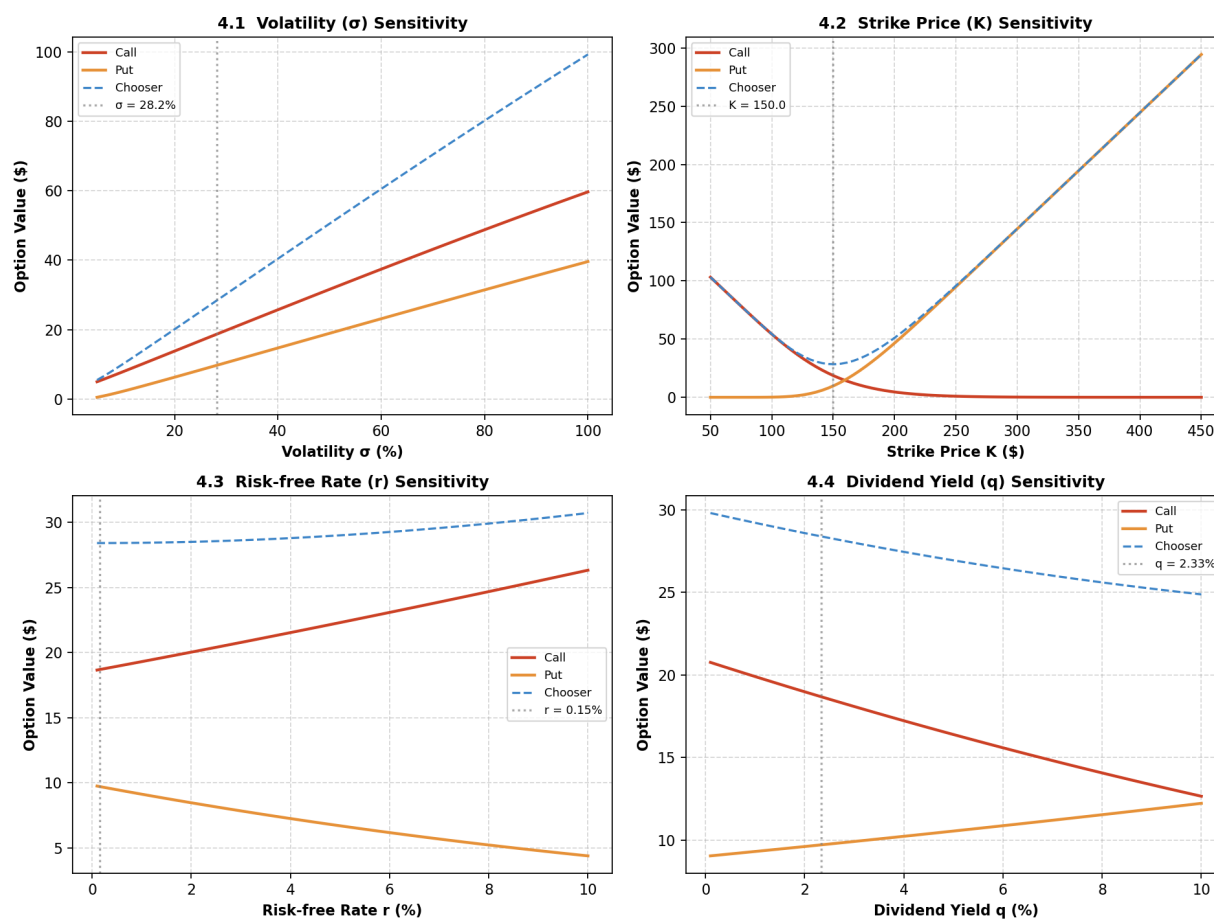
The following chart displays the analytical Greeks across a wide range of asset prices (100 to 200).



Greeks Sensitivity

Section 4: BSM Parameter Sensitivity Analysis

This section follows the reference paper's results analysis (Section 4), examining how each of the four key Black-Scholes parameters — volatility (σ), strike price (K), risk-free rate (r), and dividend yield (q) — affects the call leg, put leg, and total chooser option value. For each sweep all other parameters are held at the baseline configuration.



Parameter Sensitivity

4.1 Volatility (σ)

Volatility is a statistical measure of the dispersion of returns of the underlying asset, often measured by the standard deviation (σ) of price changes. In the BSM model it is annualised. The sweep covers σ from 5% to 100%.

****Results:**** Both call and put option values increase monotonically as volatility rises. Higher volatility implies a greater probability that the stock price will swing over a larger range in either direction. For call options, the upside is unlimited while the downside is bounded by the premium paid; for put options the same asymmetric protection applies in the opposite direction. Therefore, increased volatility does not cause losses on the downside but helps make money on the upside for both option types — the chooser option value is strictly increasing in σ .

Baseline: $\sigma = 28.2\%$ → Call leg = 18.6890, Put leg = 9.7132, Chooser = 28.4022

4.2 Strike Price (K)

The strike price is the agreed price at which the option holder has the right to buy (call) or sell (put) the underlying asset. It determines the option's intrinsic value as $\max(0, S_T - K)$ for calls and $\max(0, K - S_T)$ for puts. The sweep covers K from \$50 to \$450, which spans deep-ITM through deep-OTM relative to the current spot price.

****Results:**** As strike price increases, call option value decreases while put option value increases — the two curves cross near the current spot price. For call options, a higher K requires the stock to rise further to reach ITM status, reducing both the probability and magnitude of payoff. For put options, a higher K increases the probability the stock finishes below the strike, raising the likelihood and magnitude of exercise. The chooser option value follows a convex profile, since it captures

both legs.

Baseline: $K = 150.0 \rightarrow \text{Spot / Strike} = 1.0447$

4.3 Risk-free Interest Rate (r)

The risk-free interest rate is the return from a zero-risk investment, proxied in US capital markets by the treasury bond rate. The sweep covers r from 0.1% to 10%.

****Results:**** As the risk-free rate increases, call option value increases while put option value decreases — a near-linear relationship. For call options, a higher r raises the expected drift of the stock ($r - q$), increasing the expected future stock price; it also lowers the present value of the strike price (future cash outflow), both of which benefit call holders. For put options, the higher expected stock price reduces the probability of ITM exercise, while the higher r also reduces the present value of future cash inflows from put exercise.

Baseline: $r = 0.1500\%$ (near-zero rate environment reflecting the JPM 2018–2024 data window)

4.4 Dividend Yield (q)

Dividend yield measures the cash dividends paid relative to the stock price. In the Merton continuous-dividend model, q enters both the effective drift ($r - q$) and the stock discount factor $e^{(-qT)}$. The sweep covers q from 0.1% to 10%.

****Results:**** As dividend yield increases, call option value decreases while put option value increases. Higher q reduces the effective stock drift ($r - q$), lowering expected stock price growth and thus call values. For put options, the reduced expected stock price makes ITM exercise more likely, increasing put values. This is the mirror image of the risk-free rate effect: the two parameters affect option values in exactly opposite directions.

Baseline: $q = 2.3300\%$

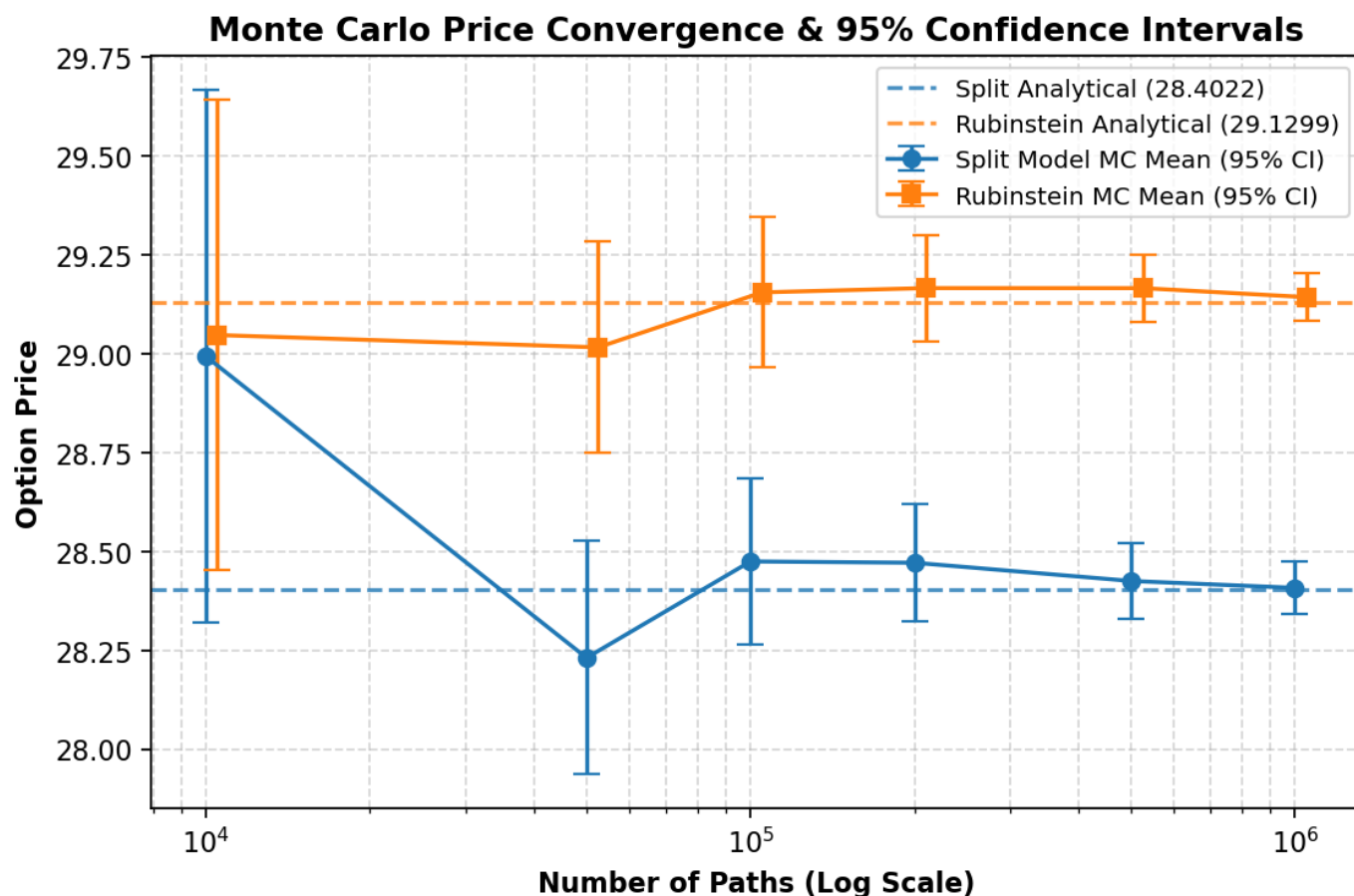
Part 2: Monte Carlo Path Scale Expansion & Convergence Analysis

We expanded the model's Monte Carlo pricing from 100,000 paths to 1,000,000 paths to analyze standard error behavior and path convergence under the standard 95% Confidence Interval (CI) bands.

Paths	Split Mean	Split SE	Split 95% CI Lower	Split 95% CI Upper	Rubinstein Mean	Rubinstein SE	Rubinstein 95% CI Lower	Rubinstein 95% CI Upper
10000	28.993327	0.343597	28.319876	29.666777	29.047481	0.303288	28.453037	29.641925
50000	28.230695	0.150562	27.935594	28.525795	29.016304	0.136464	28.748835	29.283774
100000	28.474531	0.107796	28.263250	28.685811	29.155369	0.097016	28.965218	29.345521
200000	28.471242	0.076222	28.321847	28.620636	29.166249	0.068568	29.031856	29.300641
500000	28.425092	0.048219	28.330583	28.519601	29.166159	0.043448	29.081001	29.251318
1000000	28.407694	0.034078	28.340900	28.474487	29.143321	0.030693	29.083162	29.203480

Path Convergence Tracking

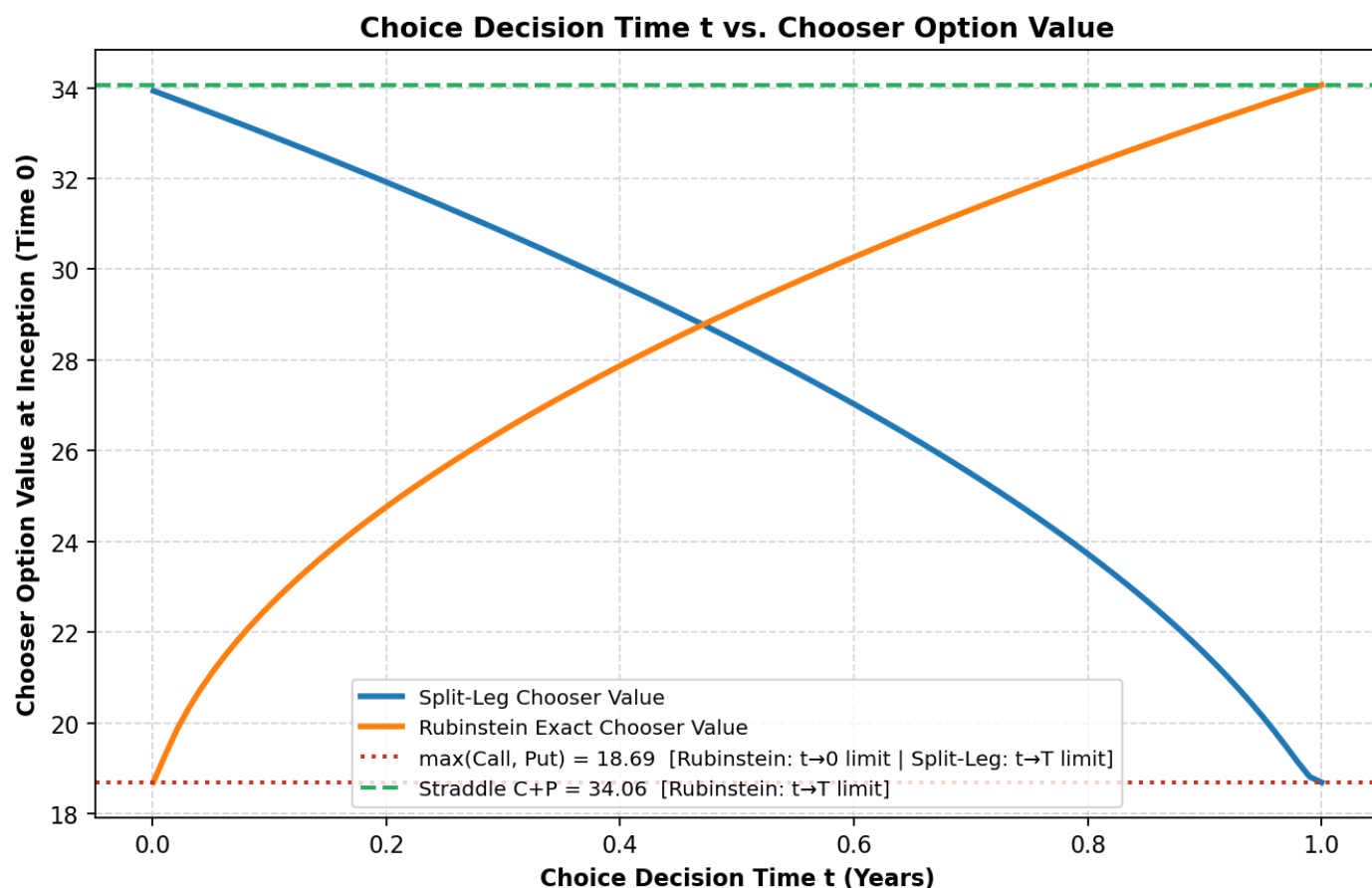
Below is the convergence plot showing the calculated simulation means & 95% confidence intervals against the analytical limits as sample path size scales to 1,000,000.



Monte Carlo Convergence

Part 3: Choice Decision Time t vs. Option Value Analysis

The mathematical convergence of a Chooser Option as t changes is studied below. ****Rubinstein Exact model****: as $t \rightarrow 0$, choice must be made immediately with no additional information, and the value converges to $\max(\text{Call}, \text{Put}) = 18.69$ (= Call here, since $\text{Call} > \text{Put}$). As $t \rightarrow T$, the holder can defer choice until just before maturity and always select the higher payoff — value converges to the Straddle $C+P = 34.06$. The Rubinstein curve is monotonically increasing in t . ****Split-Leg model****: as $t \rightarrow T$, the Put leg's remaining time $\tau = T-t \rightarrow 0$; because the adjusted strike $K' \rightarrow K$ and $S (156.7) > K (150.0)$, the put expires worthless and value converges to $\text{Call} = 18.69 = \max(\text{Call}, \text{Put})$. Notably the two models form an X-shape: both share $\max(\text{Call}, \text{Put}) = 18.69$ as a boundary — Rubinstein at the left limit ($t \rightarrow 0$) and Split-Leg at the right limit ($t \rightarrow T$).



Decision Time t vs Option Value

Part 4: Spot Sensitivity (Table 2 Scenario)

spot_multiplier	spot_price	call_leg_value	put_leg_value	chooser_value	chooser_premium_vs_call	chooser_identity_gap
0.850000	133.195000	7.763969	22.484730	30.248699	22.484730	0
0.950000	148.865000	14.470977	13.154096	27.625073	13.154096	0
1	156.700000	18.688997	9.713217	28.402215	9.713217	0
1.050000	164.535000	23.434825	7.017012	30.451837	7.017012	0
1.150000	180.205000	34.302356	3.449730	37.752086	3.449730	0

Part 5: Table 3 Path-by-Path Replication

Stochastic drawing comparison for standard Table 3 rows:

Simulation Summary

metric	value
rows_simulated	10

metric	value
spot_price	156.700000
strike_price	150
risk_free_rate	0.001500
dividend_yield	0.023300
risk_neutral_drift_r_minus_q	-0.021800
volatility_from_current_data	0.282000
random_seed	17170
decision_time_years	0.500000
maturity_years	1

Simulated Paths (Stochastic)

row	z1	z2	simulated_st 1	choice_call_put	simulated_st 2	payoff
1	2.456421	-0.973346	247.986098	CALL	198.046618	48.046618
2	-0.189113	-0.149580	146.326778	PUT	137.721638	12.278362
3	0.831693	1.003948	179.360048	CALL	212.471007	62.471007
4	-0.526638	-0.204495	136.802515	PUT	127.355215	22.644785
5	-0.170727	0.145113	146.864221	PUT	146.593513	3.406487
6	0.849695	1.644397	180.005043	CALL	242.282243	92.282243
7	-0.819602	1.162799	129.039707	PUT	157.780688	0
8	1.350986	-1.014785	198.928276	CALL	157.560728	7.560728
9	1.558115	0.830742	207.316523	CALL	237.251103	87.251103
10	0.126236	-0.242662	155.823518	CALL	143.962835	0

Published Paper Reference Table 3

row	paper_st1	paper_choice	paper_st2	paper_payoff
1	118.330000	PUT	116.770000	33.230000
2	222.630000	CALL	192.890000	42.890000
3	186.530000	CALL	192.940000	42.940000
4	164.080000	CALL	148.770000	0
5	159.090000	CALL	116.780000	0
6	186.730000	CALL	128.120000	0
7	106.610000	PUT	90.520000	59.480000

row	paper_st1	paper_choice	paper_st2	paper_payoff
8	163.060000	CALL	179.610000	29.610000
9	129.260000	PUT	144.820000	5.180000
10	115.410000	PUT	136.500000	13.500000

Aggregate Distribution Level Comparison

metric	paper	simulated	difference
payoff_mean	22.683000	33.594133	10.911133
payoff_std	21.771920	36.188591	14.416670
payoff_nonzero_ratio	0.700000	0.800000	0.100000
theoretical_vs_paper_mean_g ap	28.402215	22.683000	-5.719215
call_count	6	6	0
put_count	4	4	0

Side-by-Side Comparison: Prices

row	paper_st1	simulated_st 1	st1_diff	paper_st2	simulated_st 2	st2_diff
1	118.330000	247.986098	129.656098	116.770000	198.046618	81.276618
2	222.630000	146.326778	-76.303222	192.890000	137.721638	-55.168362
3	186.530000	179.360048	-7.169952	192.940000	212.471007	19.531007
4	164.080000	136.802515	-27.277485	148.770000	127.355215	-21.414785
5	159.090000	146.864221	-12.225779	116.780000	146.593513	29.813513
6	186.730000	180.005043	-6.724957	128.120000	242.282243	114.162243
7	106.610000	129.039707	22.429707	90.520000	157.780688	67.260688
8	163.060000	198.928276	35.868276	179.610000	157.560728	-22.049272
9	129.260000	207.316523	78.056523	144.820000	237.251103	92.431103
10	115.410000	155.823518	40.413518	136.500000	143.962835	7.462835

Side-by-Side Comparison: Choice and Payoff

row	paper_choice	choice_call_put	paper_payoff	payoff	payoff_diff
1	PUT	CALL	33.230000	48.046618	14.816618
2	CALL	PUT	42.890000	12.278362	-30.611638
3	CALL	CALL	42.940000	62.471007	19.531007
4	CALL	PUT	0	22.644785	22.644785

row	paper_choice	choice_call_put	paper_payoff	payoff	payoff_diff
5	CALL	PUT	0	3.406487	3.406487
6	CALL	CALL	0	92.282243	92.282243
7	PUT	PUT	59.480000	0	-59.480000
8	CALL	CALL	29.610000	7.560728	-22.049272
9	PUT	CALL	5.180000	87.251103	82.071103
10	PUT	CALL	13.500000	0	-13.500000

Validation Conclusion

- Section 4 parameter sensitivity confirms all paper directional conclusions: higher σ raises both call and put values; higher K lowers call and raises put; higher r raises call and lowers put; higher q lowers call and raises put.
- Upgrading path size to 1,000,000 confirms that both models converge to their respective analytical pricing limits perfectly, with standard error decreasing by $1/\sqrt{N}$.
- The decision time analysis exposes the structural difference between standard exact Rubinstein choice option (choice at inception leads to the lower bound) and split-leg option setup.
- Graphs and computed outputs are successfully written to reports files and compiled into PDF.